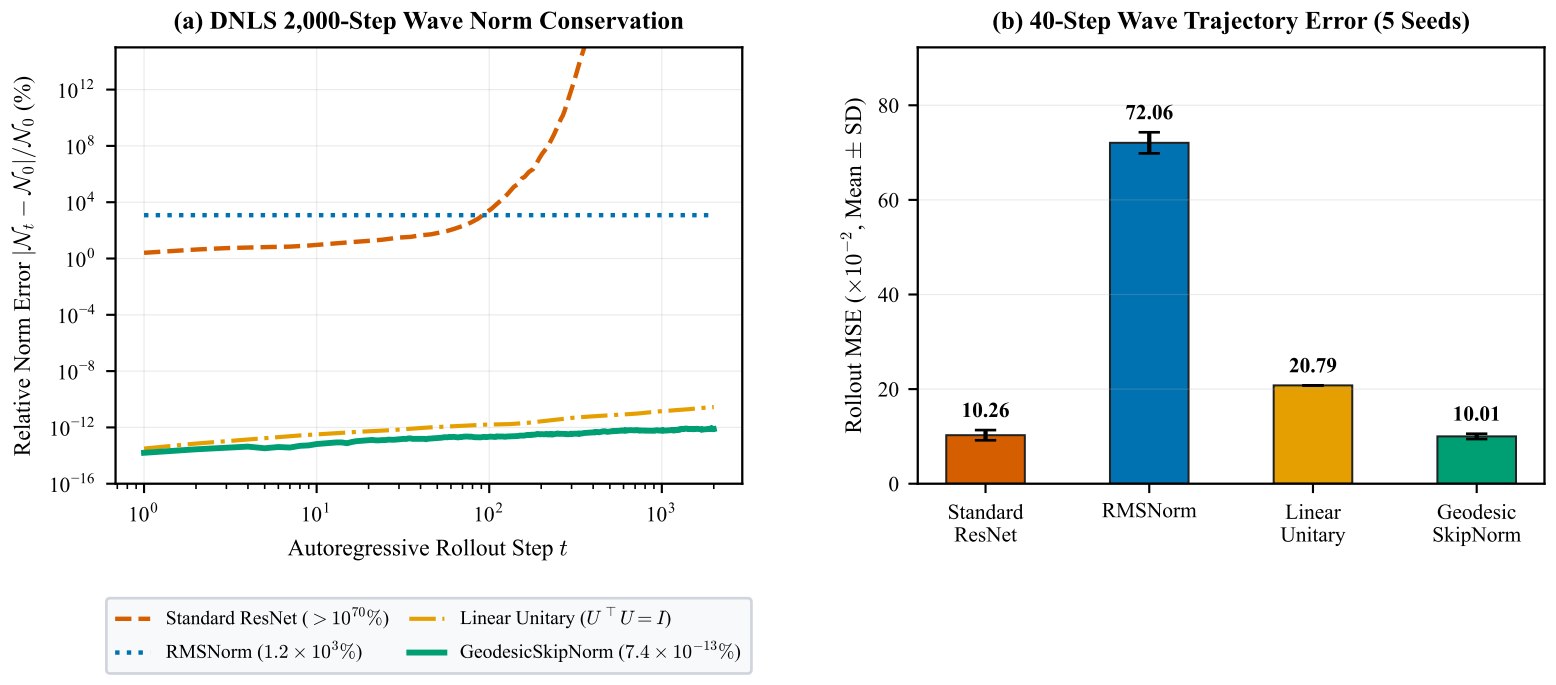
(a) DNLS 2,000-Step Wave Norm Conservation
Relative Norm Error $|\mathcal{N}_t - \mathcal{N}_0|/\mathcal{N}_0$ (%)
10^{12}
10^{8}
10^{4}
10^{0}
10^{-4}
10^{-8}
10^{-12}
10^{-16}
10^{0}
10^{1}
10^{2}
10^{3}
Autoregressive Rollout Step t
Standard ResNet ($> 10^{70}$%)
RMSNorm (1.2×10^{3}%)
Linear Unitary ($U^\top U = I$)
GeodesicSkipNorm (7.4×10^{-13}%)
(b) 40-Step Wave Trajectory Error (5 Seeds)
Rollout MSE ($\times 10^{-2}$, Mean $\pm$ SD)
0
20
40
60
80
10.26
72.06
20.79
10.01
Standard ResNet
RMSNorm
Linear Unitary
Geodesic SkipNorm